

Style transfer with LoRA adapters on Stable Diffusion 1.5

Project Work in Deep Learning

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Abstract

Two LoRA adapters are trained on the frozen UNet of Stable Diffusion 1.5 to make it learn an abstract visual mood from 30 and 35 chosen images, and we measure how well that style generalises to different prompts. 1,594,368 parameters out of 861M are updated (0.185%): each adapter weighs 6.1 MB against the ~3.4 GB of the full model. Both moods transfer clearly onto subjects foreign to their respective datasets, with CLIP deltas that are positive but small (+0.0380 and +0.0195), consistent with the fact that image-to-image CLIP similarity is dominated by content rather than style.

1 Goal

The question is whether a rank-8 LoRA, applied to the attention projections alone of a frozen UNet, can absorb something as hard to pin down as a mood from a few dozen photographs, and then apply it to prompts that share nothing with those photographs.

Everything stays frozen except the adapter: the VAE, the text encoder and the 861M parameters of the UNet. The LoRA matrices are injected into the `to_q`, `to_k`, `to_v` and `to_out.0` projections, that is, where the text conditioning enters. The style is bound to a trigger phrase that ends every training caption, and the same phrase reactivates it at inference time.

2 Method

2.1 Data

I've created one subfolder for each mood. Images are brought to 512×512 by resizing on the short side and centre cropping, and each is paired with a caption ending in its own mood's trigger. The trigger is recorded in `dataset.json` and propagated all the way to the checkpoint, so it does not have to be retyped at inference time: a suffix different from the training one leaves the adapter silently inactive, and this was the most frequent mistake in the early experiments.

run	trigger	images	character of the dataset
bright	<cjh-bright>	30	saturated, luminous, illustrative
scary	<cjh-scary>	35	desaturated, cold, foggy

Table 1: The two style datasets. The trigger is the only part shared by every caption of a mood.

2.2 Training

Both runs share the same hyperparameters, so the two adapters differ only in what they have seen: rank 8, $\alpha = 16$, learning rate 10^{-4} , 1200 steps, batch 1 with gradient accumulation 2, resolution 512×512 , seed 42.



Figure 1: Three training images per mood: **bright** on top, **scary** at the bottom.

Each step samples a random timestep, adds the corresponding noise to the latent produced by the VAE and asks the UNet to predict it, the loss is the MSE between predicted and true noise, and the gradients reach the LoRA matrices only. Gradient checkpointing and fp16 are what make the training fit within the 6 GB of the GPU used (GTX 1060), on this Pascal architecture, fp16 saves memory more than it speeds things up.

trainable parameters	1,594,368
total UNet parameters	861,115,332
fraction trained	0.185 %
size of the adapter	6.1 MB
size of the full model	~3.4 GB
GPU	GTX 1060 6 GB
duration of one run	1h 27m

Table 2: Cost of the fine-tuning.

2.3 Evaluation

The 11 evaluation prompts describe quite different subjects: four neutral with respect to both moods, three ambiguous, and two per mood chosen to pull the result one way. The seed depends only on the prompt index and not on the variant, so base and adapter start from the same noise and the differences are attributable to the adapter alone.

The metric is the mean CLIP similarity (`clip-vit-base-patch32`) between the generated images and the mood’s reference dataset. The absolute value is not interpretable on its own, so what gets reported is the delta against the same images generated by the base model.

3 Results

run	base	lora	delta	std	prompts improved
bright	0.5877	0.6257	+0.0380	0.0673	7 / 11
scary	0.5571	0.5766	+0.0195	0.0643	8 / 11

Table 3: Mean CLIP similarity against the reference dataset of each mood.

Both deltas are positive and both are small, which is the expected outcome and not a disappointing result: image-to-image CLIP similarity is dominated by the subject, and the subject is held fixed across the columns by construction. The metric should be read as weak evidence; the comparison grid remains the primary evaluation.

On visual inspection the transfer is already clear at scale 1.0, so no sweep over the adapter strength was needed. **bright** turns the photographs into illustration: the lighthouse acquires painted rock and a poster-like sky, the fisherman becomes a painterly portrait, the waterfall shifts towards turquoise and violet. **scary**, by contrast, works by subtraction: it removes saturation, cools the palette and fills the middle ground with haze, to the point that the forest turns yellow-grey and the wolf in the snow loses almost all of its colour. It is worth noting that the scary result show just a change in the mood, rather than a complete transformation (like adding blood, or horror subjects).

Focusing on the rows of the neutral subjects, we can see that even if those are have nothing to do with either dataset, they are still restyled coherently, showing a generalizationism of these modules.

The main problem I can see in this project is that the trigger carries meaning: the CLIP tokenizer breaks the trigger into ordinary pieces ['in', '<', 'cj', 'h', '-', 'scary', '>', 'style'] and *scary* and *bright* are English words the base model already understands and might influence the model's behavior.

A Comparison grids

Every row uses the same seed across the three columns: the differences are attributable to the adapter alone. The adapters are applied at scale 1.0.



Figure 2: Subjects neutral with respect to both datasets: they measure how far the style applies outside its own domain.

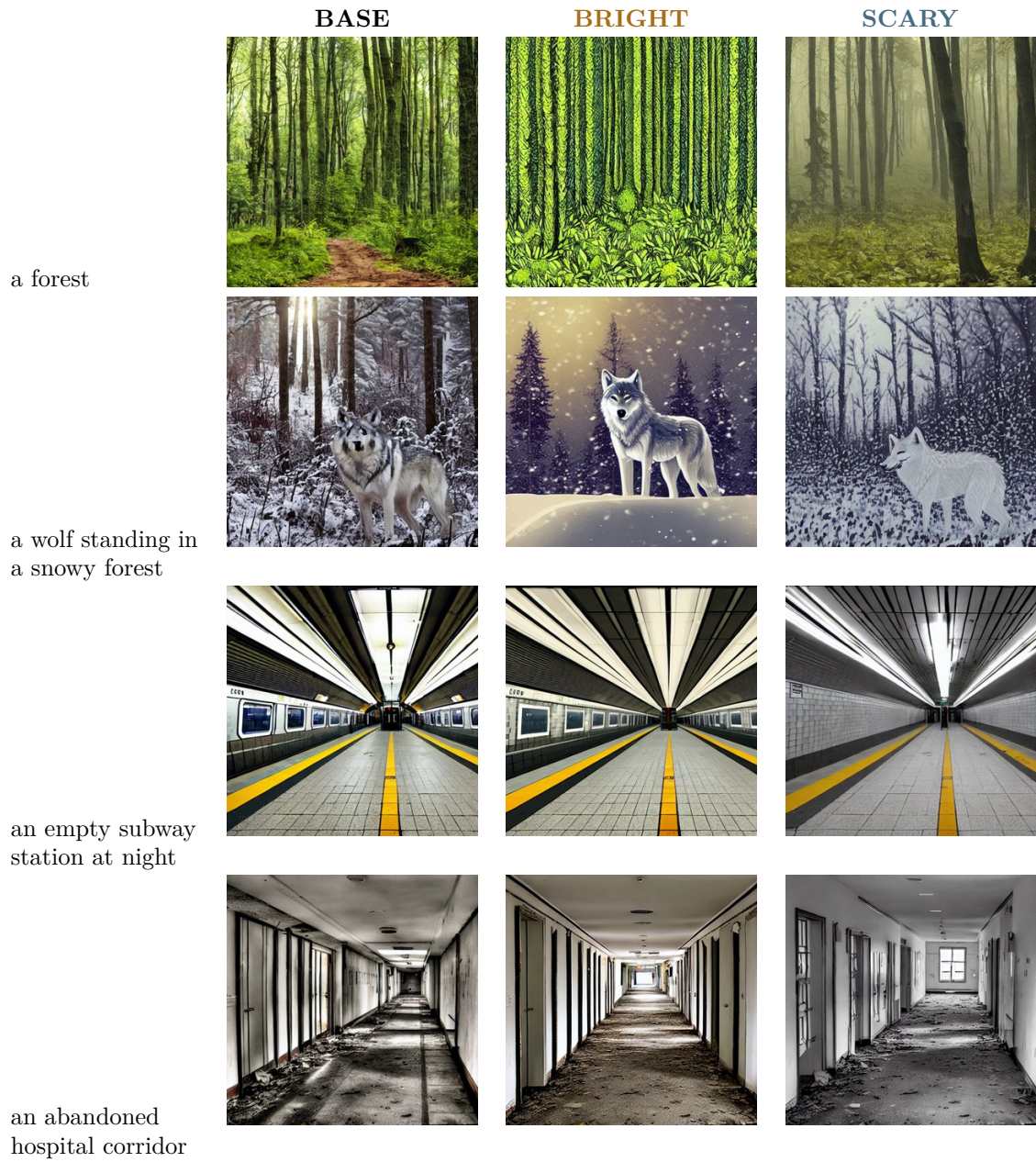


Figure 3: Ambiguous subjects: it is the adapter’s mood that decides which way they lean.



Figure 4: Subjects that push towards one mood or the other: the first towards **scary**, the other two towards **bright**.